

Implementation Status Report

AI-Powered Industrial Vibration Intelligence Platform

A consolidated record of everything built to date — the condition-monitoring feature set, the signal-processing engine, the data model, and the API and interface layers that carry a vibration capture from an edge device through to a graded machine-health verdict.

Report date

7 September 2026

Branch

laxman-dev

Codebase

42,877 lines

Status

Active development

01 Executive Summary

SensoVibe is a full-stack condition-monitoring platform for rotating machinery. It ingests vibration captures from edge DAQ devices, transforms them into diagnostic spectra and scalar health features, grades those features against configurable thresholds, and presents the result through an industrial analysis workspace.

99 API ENDPOINTS	25 DATABASE TABLES	18 MIGRATIONS	13 API ROUTERS
75 BACKEND MODULES	232 FRONTEND MODULES	10 HEALTH FEATURES	45 DELIVERED FEATURES

What the platform does end to end

An edge device polls the backend for its acquisition plan — sample rate, lines of resolution, window, averaging, overlap and channel map — captures a block, and posts it back. The backend parses the payload, computes every enabled plot type, extracts ten scalar features per channel, grades each against its threshold rule, stores the derived artefacts, and fires webhooks on alarm. The frontend then renders the capture across the analysis workspace, comparing it against a saved baseline.

capture → parse (CSV / PDF) → per-channel arrays → DSP (6 plot types)
→ feature extraction (10 codes) → threshold grading → persist + fingerprint cache
→ webhook dispatch on alarm → analysis workspace

Scope note. This report covers implemented, reachable behaviour. Features that exist as scaffolding or are declared but not wired are listed separately in Section 8 rather than counted as delivered.

02 System Architecture

LAYER	TECHNOLOGY	RESPONSIBILITY
Edge device	UDP / MQTT collector	Polls acquisition config by device ID; captures and posts raw blocks.
API	FastAPI 0.115 · Uvicorn	13 routers, JWT-guarded, role-scoped write access.
Services	NumPy 2 · SciPy 1.14	DSP engine, feature extraction, threshold grading, storage orchestration.
Persistence	PostgreSQL · SQLAlchemy 2 · Alembic	25 tables; derived artefacts cached by config fingerprint.
Client	React 18 · TypeScript · Vite	Analysis workspace, equipment master, settings, dashboard.
Charting	ECharts 6 · echarts-gl · Plotly	2D diagnostic charts plus 3D waterfall viewport.
State	TanStack Query 5 · Axios	Server-state cache; silent token refresh with request queueing.
Auth	python-jose · passlib/bcrypt	JWT access tokens with rotating opaque refresh tokens.
Parsing	pdfplumber · Pillow	PDF measurement reports and equipment imagery.

03 Feature Inventory

Every feature below is implemented and reachable in the running application.

Authentication & access control

ID	FEATURE	IMPLEMENTATION
F-01	Email/password login issuing JWT access plus opaque refresh tokens	POST /auth/login, /token
F-02	Silent token refresh with concurrent-request queueing	axios interceptor + /auth/refresh
F-03	Logout with server-side refresh-token revocation	POST /auth/logout
F-04	Session bootstrap from sessionStorage on reload	GET /auth/me
F-05	Role-guarded routes, nav items and write operations	ProtectedRoute, require_write_access
F-06	User administration: create, edit, reset password, delete, role assignment	/users, /roles

Asset & plant master data

ID	FEATURE	IMPLEMENTATION
F-07	Equipment register with search, type and criticality filters, pagination	GET /equipment/
F-08	Six-step equipment wizard with live completeness scoring	EquipmentForm.tsx
F-09	Equipment image upload, fetch and delete	/equipment/{id}/image
F-10	Sensor configuration CRUD nested under equipment	/equipment/{id}/sensors
F-11	Plant, Area and Line hierarchy with cascade delete	/plants, /areas, /lines
F-12	AI-readiness scoring: five checks resolved to a percentage	/equipment/{id}/ai-readiness
F-13	Lookup catalogue: 18 static lists plus a derived plant list	/lookups/
F-14	Rotating-components and operating-process specification tabs	RotatingComponentsTab.tsx

Acquisition & ingestion

ID	FEATURE	IMPLEMENTATION
F-15	Edge acquisition JSON per device ID: rate, LOR, window, averaging, overlap, channel map	GET /acquisition/config/{device_id}
F-16	Device measurement ingest endpoint	POST /ingest/measurements
F-17	Raw waveform snapshot ingest, stored per channel	POST /ingest/raw
F-18	CSV and PDF upload with synchronous parse, plot and feature pipeline	POST /measurements/upload
F-19	Sample-rate and timestamp validation with unit-scale inference	raw_vibration.py
F-20	Plot configuration upsert per sensor	/measurements/configure/{sensor_id}

Analysis & diagnostics

ID	FEATURE	IMPLEMENTATION
F-21	Capture timeline browser with date-range filter and day chips	GET /measurements/uploads
F-22	Six computed plot types with zoom, pan, crosshair, threshold overlays, PNG export, fullscreen	/uploads/{id}/plots
F-23	Ten-feature health dashboard with summary cards and grouped status tables	/uploads/{id}/features
F-24	Feature comparison against baseline with percentage difference	/uploads/{id}/features/compare
F-25	Per-feature 32-segment trend cards	/uploads/{id}/factor-trends
F-26	3D FFT waterfall across many captures, with spectral peak detection	GET /measurements/waterfall
F-27	Casing orbit plot using frequency-domain double integration to displacement	/uploads/{id}/orbit
F-28	1X amplitude and phase migration, plotted polar across captures	GET /measurements/one-x-migration
F-29	Vibration vector analysis: per-block amplitude and de-rotated phase with drift detection	/uploads/{id}/vector
F-30	Raw waveform tab: snapshot list, windowed trace, statistics and spectrum	/raw/snapshots, /raw/latest
F-31	Statistics panel: RMS, peak, peak-to-peak, crest, kurtosis (both conventions), skewness	raw_analysis.py
F-32	FFT cascade plot	FFTCascadePlot.tsx

Baselines & thresholds

ID	FEATURE	IMPLEMENTATION
F-33	Baseline management: list, search, filter, set primary, load for analysis	/baselines
F-34	Create a baseline from an existing upload, copying plots and features	POST /baselines/from-upload/{id}
F-35	Direct baseline file upload	POST /baselines/upload
F-36	Threshold rule engine: 5 rule types over 10 feature codes, factory defaults, per-channel scope	/thresholds/rules, /defaults
F-37	Reset any rule to its factory default	POST /thresholds/rules/{id}/reset
F-38	Threshold limits shown beside each feature status in the Health tab	FeatureThresholdLimitsSection.tsx

Platform, integrations & interface

ID	FEATURE	IMPLEMENTATION
F-39	API key issuance, listing, revocation and deletion	/integrations/api-keys
F-40	Webhook CRUD with secret rotation, test dispatch and delivery log	/integrations/webhooks

ID	FEATURE	IMPLEMENTATION
F-41	Alarm-driven webhook notification carrying equipment context	webhook_service.py
F-42	Operations dashboard with live fleet health, alerts feed and analytics feed	GET /dashboard/summary
F-43	Platform settings module for acquisition defaults administration	AcquisitionSection.tsx
F-44	Plant selector filtering all data plant-wise	TopNav.tsx
F-45	Golden-ratio layout system, responsive breakpoints, toasts, collapsible sidebar	components/layout, components/ui

04 Signal Processing Engine

The DSP core was rebuilt to follow a single reference specification, so every spectrum in the platform comes from one implementation with one set of conventions.

Core formulas

QUANTITY	DEFINITION	NOTES
FFT block size	$N_{\text{FFT}} = \max(16, \text{LOR})$	LOR sizes the block; it is not a line count.
Line count	$N_{\text{FFT}} / 2 + 1$	DC and Nyquist both included.
Resolution	$\Delta f = F_s / N_{\text{FFT}}$	Axis $f_k = k \cdot \Delta f$.
Coherent gain	$\text{CG} = \text{mean}(w)$	Amplitude correction, not noise power.
Magnitude	$A[k] = 2 X[k] / (N_{\text{FFT}} \cdot \text{CG})$	DC and Nyquist are not doubled.
Averaging	$\text{step} = \max(1, \text{round}(N \cdot (1 - \text{overlap})))$	Linear magnitude averaging across frames.
Envelope	band-pass 2–5 kHz → Hilbert → Hann block	Demodulated bearing envelope.
gE envelope	rectify → 1-pole low-pass 500 Hz	Scalar bearing feature.
Velocity RMS	$\text{detrend}(f \text{ a dt}) \times 1000$	Trapezoidal, time domain, mm/s.

Windows implemented

Rectangular, Hann, Hamming and the 5-term ISO 18431-1 flat-top, each with its own coherent-gain correction so amplitude stays true regardless of window choice. Verified: a 2.0 g bin-centred sine returns **2.0000000 g**, and off-bin loss is pure Hann scalloping at −0.224 dB, inside the −1.42 dB worst case.

Axis contract

Every plot carries a machine-readable axis descriptor rather than free-text labels alone: units, an explicit linear-scale declaration, the amplitude convention, and DC/Nyquist inclusion flags. A frequency axis reports `includes_nyquist: false` honestly when an Fmax limit clips the top of the axis.

GRAPH	X AXIS	Y AXIS	AMPLITUDE
Time waveform	seconds, relative	engineering unit	0-to-peak
FFT spectrum	Hz, DC to Nyquist	engineering unit	0-to-peak
Envelope spectrum	Hz, DC to Nyquist	engineering unit	0-to-peak
Envelope waveform	seconds, i/Fs	engineering unit	0-to-peak
Circular waveform	amplitude	amplitude	0-to-peak
Trend plot	seconds, within capture	RMS	RMS
3D waterfall	Hz	capture datetime, non-uniform	0-to-peak

Convention change. LOR now sizes the FFT block directly, which doubles Δf for the same LOR compared with the earlier `N = 2 × LOR` behaviour. The plot cache is version-gated and regenerates automatically; stored feature rows are not version-gated and should be recomputed before old and new rows are trended together.

05 Health Features & Grading

Ten scalar features are extracted per channel per capture, each graded by its own rule type.

FEATURE	RULE TYPE	WARNING	CRITICAL	INDICATES
rms	absolute max	0.01	0.02	Overall energy
peak	absolute max	0.05	0.10	Unsigned maximum
crest_factor	range	3.0	5.0	Impulsiveness
kurtosis	absolute max	3.5	5.0	Excess; impacting
fft_band_energy_0_500	percent baseline	120%	150%	Low-band energy
amplitude_1x	percent RMS	20%	40%	Unbalance
amplitude_2x	percent RMS	10%	20%	Misalignment
amplitude_3x	percent RMS	5%	15%	Looseness
envelope_rms	percent baseline	100%	125%	Bearing defect
noise_floor	absolute dB	-60	-54	Broadband floor

Rules are editable per sensor and per channel through the Settings module, and any rule can be reset to its factory default. Grading results drive the Health tab status badges, the fleet dashboard and alarm webhook dispatch.

06 Data Model

25 tables built up across 18 Alembic migrations.

DOMAIN	TABLES
Identity	users, roles, user_roles, refresh_tokens
Plant hierarchy	plants, areas, lines
Assets	equipment_masters, sensor_configurations
Acquisition	plot_configurations, sensor_data_uploads, measurement_upload_data
Derived artefacts	plot_results, measurement_channel_features, measurement_channel_feature_trends
Raw capture	raw_vibration_captures, raw_vibration_channels
Baselines	sensor_baselines, baseline_plot_results, baseline_channel_features
Grading	feature_definitions, feature_threshold_rules
Integrations	api_keys, webhooks, webhook_deliveries

Caching strategy. Derived plots are keyed by a SHA-256 fingerprint of the acquisition configuration and the DSP algorithm version. Changing sample rate, LOR, Fmax, data type, enabled plots or the algorithm itself invalidates the cache automatically, so a stored plot can never disagree with the settings that produced it.

07 API Surface

ROUTER	ENDPOINTS	RESPONSIBILITY
auth	5	Login, token, refresh, logout, current user
users	7	User administration and role catalogue
equipment	12	Equipment CRUD, image, sensors, AI readiness
plants	5	Plant, area and line hierarchy
lookups	3	Dropdown catalogues
measurements	20	Upload, plots, features, trends, waterfall, orbit, vector, migration, raw
acquisition	2	Edge device configuration by device ID
ingest	2	Device measurement and raw snapshot ingest
baselines	8	Baseline lifecycle, plots and features
thresholds	8	Rule CRUD, defaults and reset
integrations	13	API keys and webhooks
dashboard	1	Fleet health summary

08 Current Status & Open Items

Not yet implemented

ITEM	STATE	DETAIL
Change password	PENDING	UI renders a placeholder; no backend endpoint exists.
Per-user plant scoping	PENDING	Auth service returns an empty plant list, so every user sees every plant.
Stacked-waveform waterfall	PENDING	Stacked spectra exist; the waveform variant does not.
STFT waterfall	PENDING	The current waterfall stacks one spectrum per capture, not frames within one capture.
Global search	PENDING	Input rendered with no submit handler.
Trend Analysis tab	SUPERSEDED	Factor trends were moved into the Health tab.

Known follow-ups

- **Feature recomputation.** Stored feature rows have no algorithm-version gate, so rows computed before the DSP convention change will not regenerate on their own.
- **Envelope waveform rendering.** The plot type is computed and served; the frontend renderer is still to be wired.
- **Noise floor units.** The only dB-valued quantity in the system, sitting outside the otherwise strictly linear axis convention.
- **Dormant components.** Several built panels — asset intelligence, completeness engine, historical trend hook — are functional but not in the active render tree.

Verification performed

The DSP engine has been validated against the reference specification's own truth tables: sample-rate parsing, block sizing, frequency resolution, coherent gain for all four windows, single-sided amplitude scaling, zero-padding attenuation, overlap and averaging budgets, envelope block sizing, velocity integration, and the full axis contract across all six plot types — all passing.